[Market Impact and adoption]

A Unified Power Management Solution for AI Data Centers and Legacy Data Centers

Shuming Zhu

Solution Architect , Lattice/AMI

OPEN Compute Project SOLUTION PROVIDER®

OCP PLATINUM® Member

The presentation will focus on challenges of power management that arise as traditional data centers evolve into AI data centers.

Agenda

- New Data Center Power Architecture & Device Challenges
- Extending Power Management from 9 Dimensions to Support AI Data Centers
- Wrap-up and Looking-ahead
- Q&A

Evolving Power Architectures in Data Centers



PDU with AC output



Power shelf with ~48 V_{DC} output
(ORV3/ORV3 HPR)



+/- 400 V_{DC}/800 V_{DC} Power Rack
(HPR V4)

*Photo source: Internet

New Devices in AI Data Centers

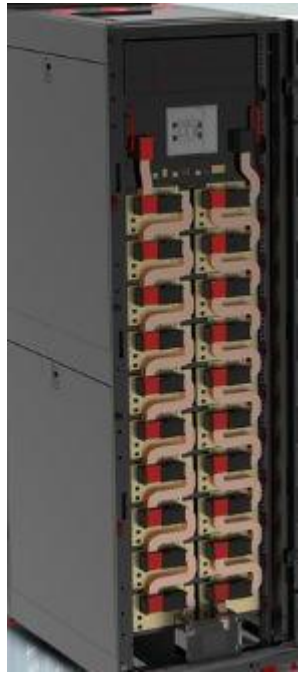
CDU



OCP Open Rack Power shelf



Super Capacitors/OCP CBU



Rack Manager/OpenRMC



OAM_eFuse

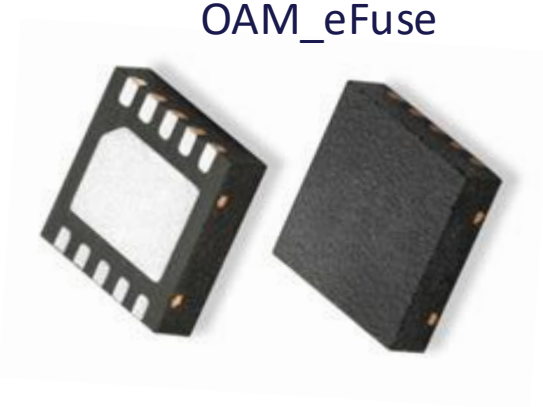


*Photo source: Internet

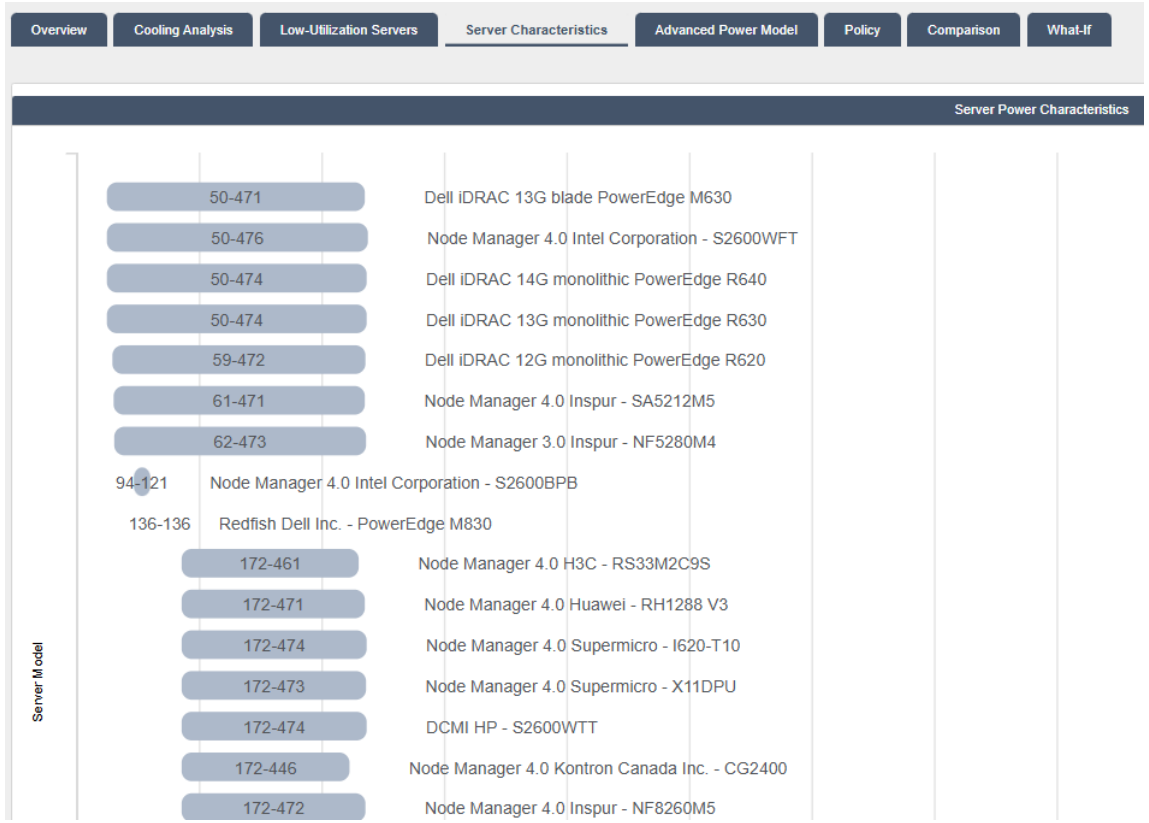
Data Center Power Management 9x Dimensions

- Power Capacity planning
- Real time Power data/metrics/telemetry
  - Current /Voltage /Power /Phase/
- Power Capping
- Emergency Power Reduction
- Carbon tracking per job
- Real time PUE
- Event Handling
- Communication Protocol
- Power on sequence

Capacity planning

GPU servers are changing the data center power planning.

a. stranded power
b. over-subscription



Power | Space| Weight| Thermal |Network

↓ ↑

Power planning

↓ ↑

IT power vs. non-IT power

↓ ↑

Rated Power(static) vs. Projected Power(Dynamic)

Rated Power(static) vs. Projected Power(Dynamic)

Real-time Power Monitoring(metrics/telemetry)

Legacy Data centers

Power



VA



AC Current



AC Voltage



Phase



Power Factor



Legacy/AI Data centers

Power



VA



AC Current



AC Voltage



Phase



Power Factor



DC Current



DC Voltage



eFuse

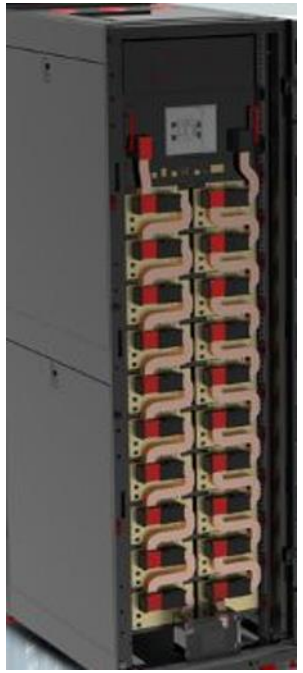


EP



Power Sampling Interval

AI data centers require smaller power sampling interval than legacy data centers, WHY?

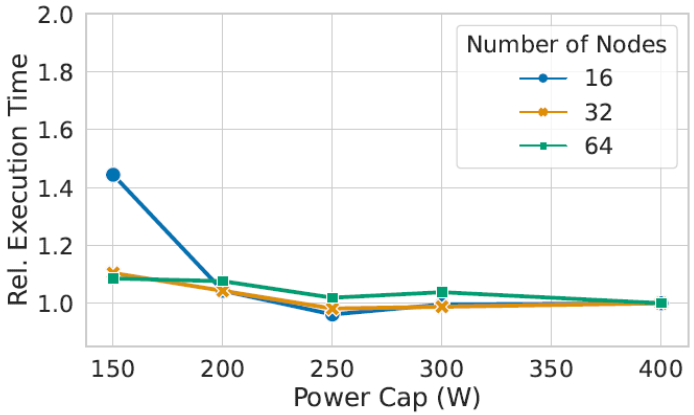


- The power variation of GPU servers is larger than that of general-purpose server, a small sampling interval helps enable continuous feedback and dynamic adjustment of power allocations.

- New devices in AI data centers , e.g. Super Capacitors/OCP CBU deliver huge bursts of current for seconds, not minutes.

- The sampling interval at a lower hierarchy level should be shorter than that at a higher hierarchy level.

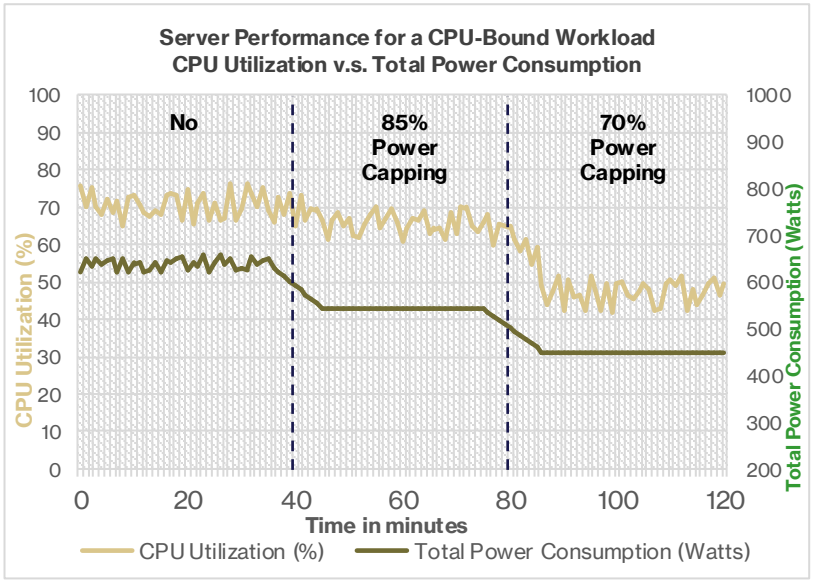
Power Capping(GPU&CPU)



(f) Spectrum - Medium

* “Our investigation shows that applying a power cap of up to 50% of the A100 GPU’s TDP to MILC jobs results in less than 15% performance penalty.”

Fatih Acun, Zhengji Zhao, Brian Austin, Ayse K. Coskun, Nicholas J. Wright
“Analysis of Power Consumption and GPU Power Capping for MILC”



A Case When Power Capping Can Reduce Power by 15% with Negligible Performance Impact for a CPU-bound Workload That is Not Fully Utilizing System Resources.

1-node, 2x production Intel Xeon 4th gen Scalable Processors (48 cores) on Intel - D50DNP, DDR5 memory total 256GB (16x 16GB), , HT On, Turbo On, SNC On, CentOS Linux release 7.9.2009, 6.1.0-rc2, One API HPC Toolkit 2023.0.0 , Intel® oneAPI Math Kernel Library (version 2023.0.0), LAMMPS update 2 for Stable release 23 June 2022, RAPL

Tiering Power Capping

Policies based on Power predictions *

Data Center Level

Cluster/group Level

Rack Level(Computing Trays + Network trays)

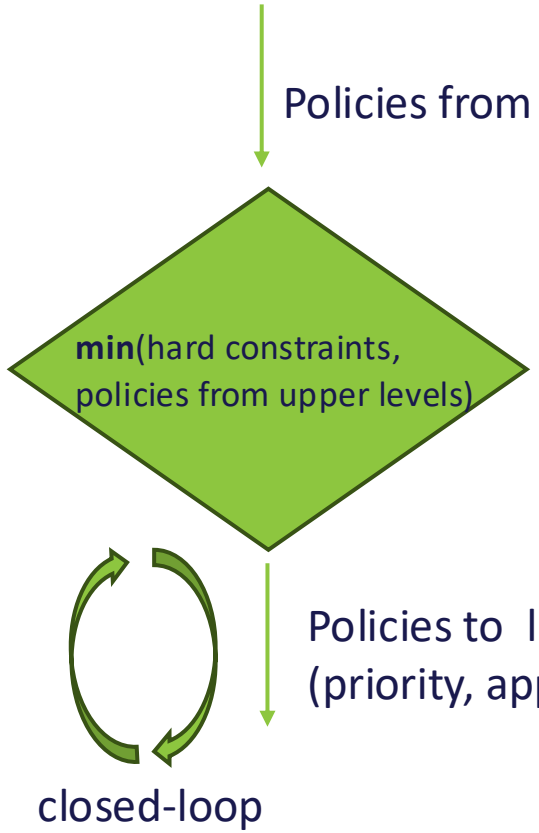
System level(GPUs + CPUs)

Component level(GPUs, CPUs)

Data

Execution

(e.g. OpenRMC)



*AI models examples : LightGBM regression; LSTM; XGBOOST; Random Forest;

Emergency Power Reduction

A real case

8:56 AM Dec 18, Aisle temp alerted from Room C HK DC	Detected issues from IT environment
9:01 AM Temp of multiple sub-room in Room C increased dramatically, found chiller abnormal.	
9:09 AM Switched 4+4 active/standby chillers, failed	
9:17 AM Triggered emergency plan: auxiliary heat dissipation, emergency ventilation, server began to halt	server can endure 20min w/o CRAC
10:30 AM Reduced workload of the entire data center	Reducing WL can't stop worsening (20-80 rule)
12:30 Chiller vendor arrived, checked cooling tower, water pipeline, chiller condenser, shutdown server protectively	Vendors need 4h for onsite support
14:47mandatory fire sprinkler triggered due to high temp	No critical temperature No EPR(Emergency Power Reduction)
15:20 Unlocked group control of chillers, 1st chiller recovered, temp dropped	Solved the issue manually
18:55 4 chiller recovered	
19:02 started servers	
00:30 Dec 19 all service recovered	

- Soft EPR(via OpenBMC)
  - CPU min Power
  - GPU capped to min. Power
  - Fan at the lowest speed
  - switch off non-critical outlets
- Hard EPR
  - Turn off all servers
  - Turn off Open Rack Power shelves/PDUs

Real Time PUE



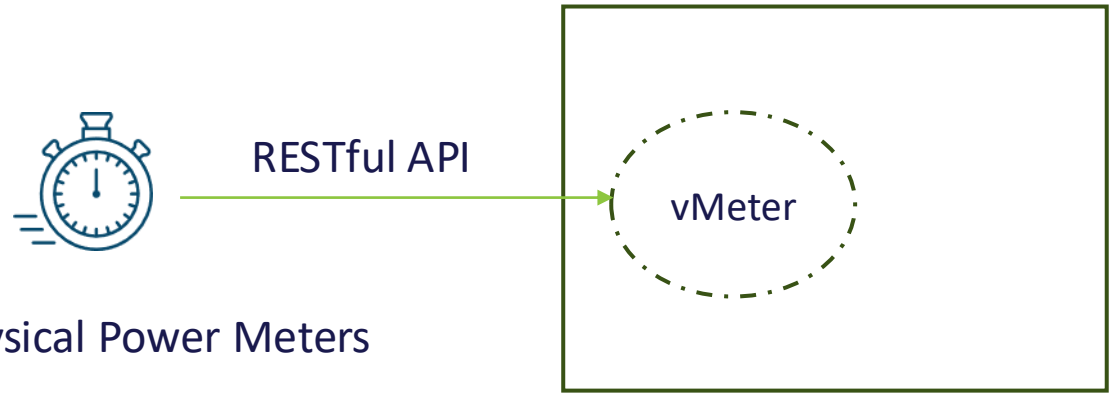
Definition:

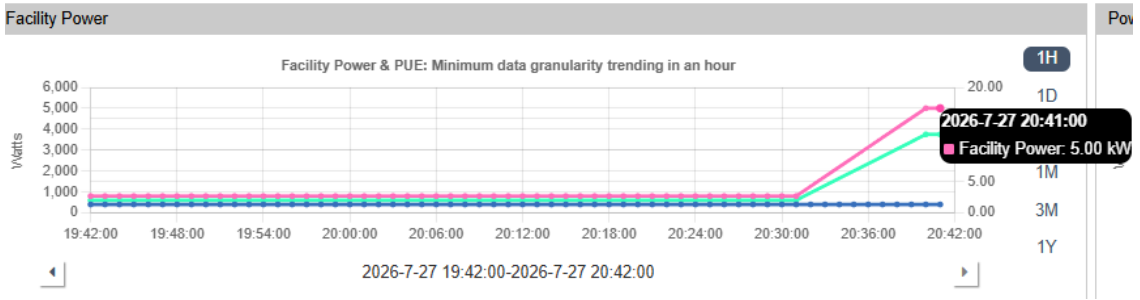
$$PUE = \frac{Total\ Facility\ Power}{IT\ Equipment\ Power};\ 1 \leq PUE$$

Challenges:

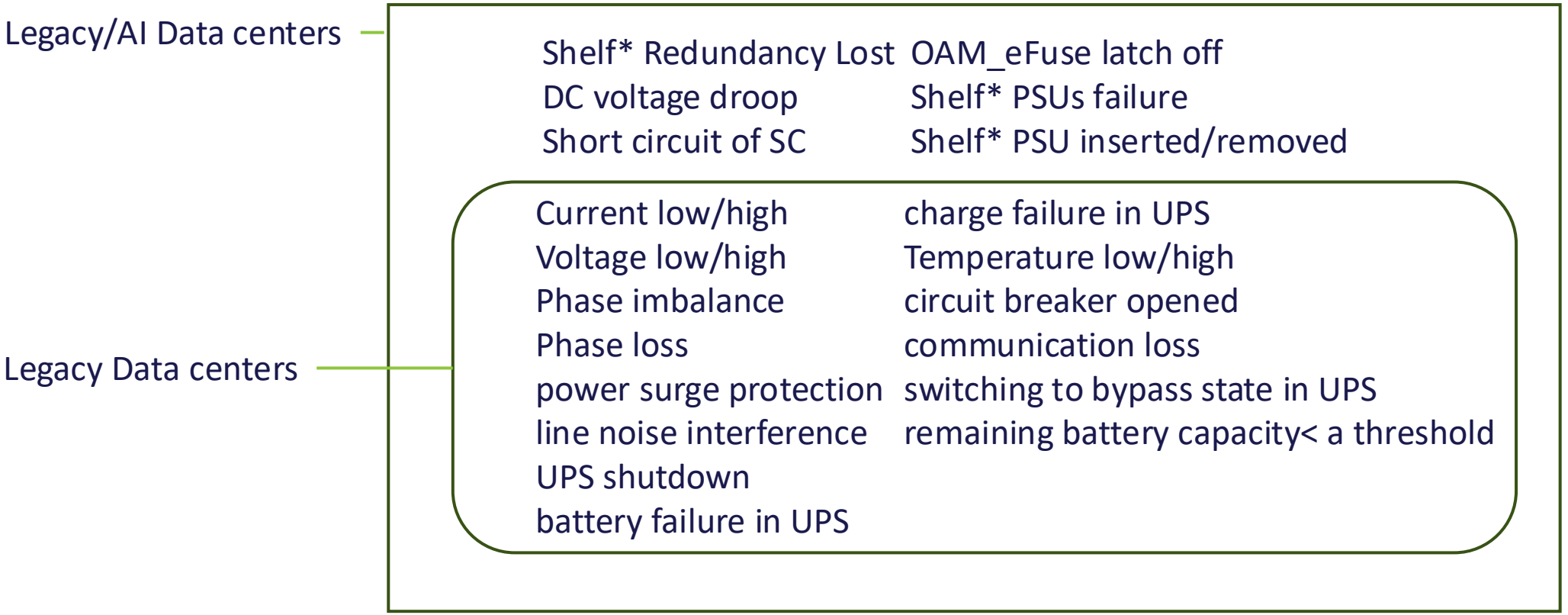
In the AI era, PUE remains a valuable metric, especially real-time PUE. The challenge is building a solution that can support a wide range of smart meters with a variety of protocols.

Solutions:





Event Handling

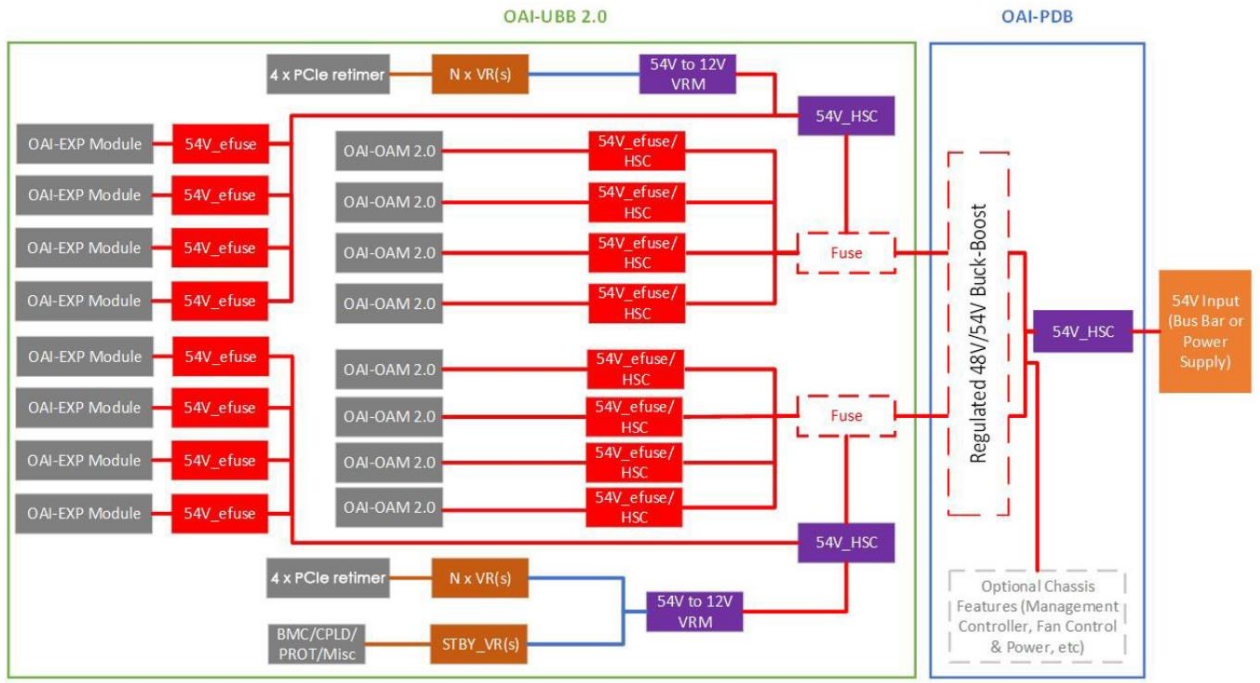


*OCP Open Rack Power shelf

OAM_eFuse management

- Resetting via the Reset pin/command rather than full power cycling prevents connector arcing and system-level disturbances.

- Standard SMBAlert# Path (PMBUS compliant), The eFuse sets STATUS bits to notify the FPGA/BMC that it needs to release SMBAlert#.



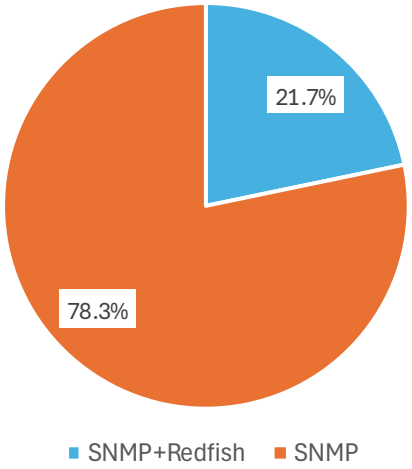
Changes of Communication Protocols

In the AI era, SNMP protocol support is still essential for power devices.

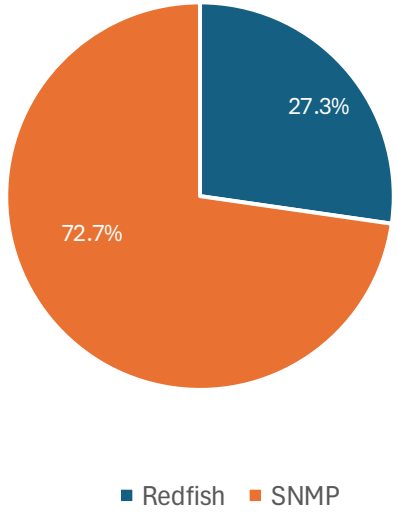




PDU Communication Protocol*



Open Rack Power Shelf Communication Protocol*



*based on PDU/OCP Open Rack Power shelf brands on open market; 11 power shelf brands; 23 PDU brands.

Power on sequence

- To avoid excessive inrush current, it's necessary to power on compute racks by multiple stages.
- A fine-grained Power on sequence of compute racks should be executed by the rack-level manager.
- Manage storage power on/off sequence by the policy within the group, e.g. Lustre file system (network->JBODs->MDS/MDT->OSS/OST->LNET->Clients)

	Power On order
1	UPS / Capacitor racks
2	Power distribution racks
3	Cooling racks
4	Networking racks
5	Storage racks
6	Compute Racks(grp 1, 2,3...)

	Power On Compute racks by steps
1	Group 1
2	Group 2
3	Group 3
4	Group 4
5	Group 5
6	...

Wrap-up and looking ahead

- Data center power management is a long journey that evolves alongside the data center itself
- End users require a unified power management solution that works for both legacy data centers and AI data centers.
- Unlike hyper-scalers, who can largely develop power management strategies on their own, enterprises need more support from the community.
- A data center power management committee can help by:
  - Translating best practices from hyper-scalers to enterprises
  - Improving energy efficiency at different levels, especially the rack level
  - Influencing silicon and component vendors from the design stage

Embrace change. Lead the future!



*Courtesy of Yong Yu

- Contact us here